

What is computational thinking?

Computational thinking is the collective thought process involved in figuring out problems and finding their solutions in ways that can be understood by a computer, or human, or both.

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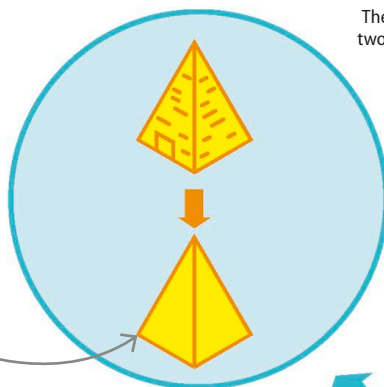
Problem solving

The goal of computational thinking is to produce instructions that enable a computer to solve a particular problem. These instructions have to be written in a language that computers understand, letting them know what they can do, and how. There are four main stages to computational thinking.

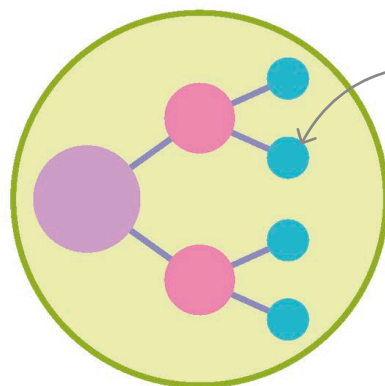
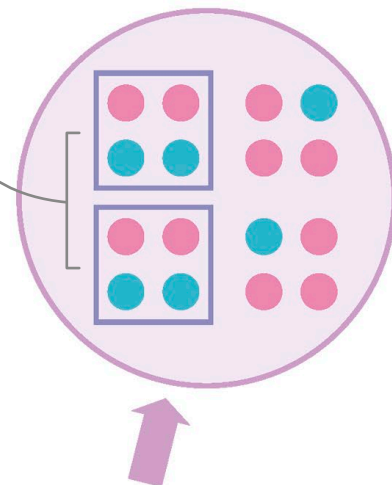
2. Abstraction

In order to write efficient sets of instructions, computer scientists need to be able to look at a situation, or problem, and work out its essential and non-essential parts. An abstraction is a model of a system, or object, that leaves out unnecessary details.

The lower pyramid is an abstraction of the one above, with details like doors and bricks removed.



The patterns in the two squares match.



Breaking down a large problem into smaller ones is the first step in computational thinking.

1. Decomposition

Decomposition is the process of breaking down a problem into smaller sub-problems. It's usually possible to break down an overwhelming task into several smaller, more manageable tasks. Tackling each of these one-by-one gets the original problem solved.

3. Pattern recognition

While solving a problem, computer scientists try to find parts that are similar to problems they've solved before. Recognizing these patterns is useful as it means that they can use or modify an existing solution to solve their current problem.

